Lecture 4

Investment and Capital Accumulation

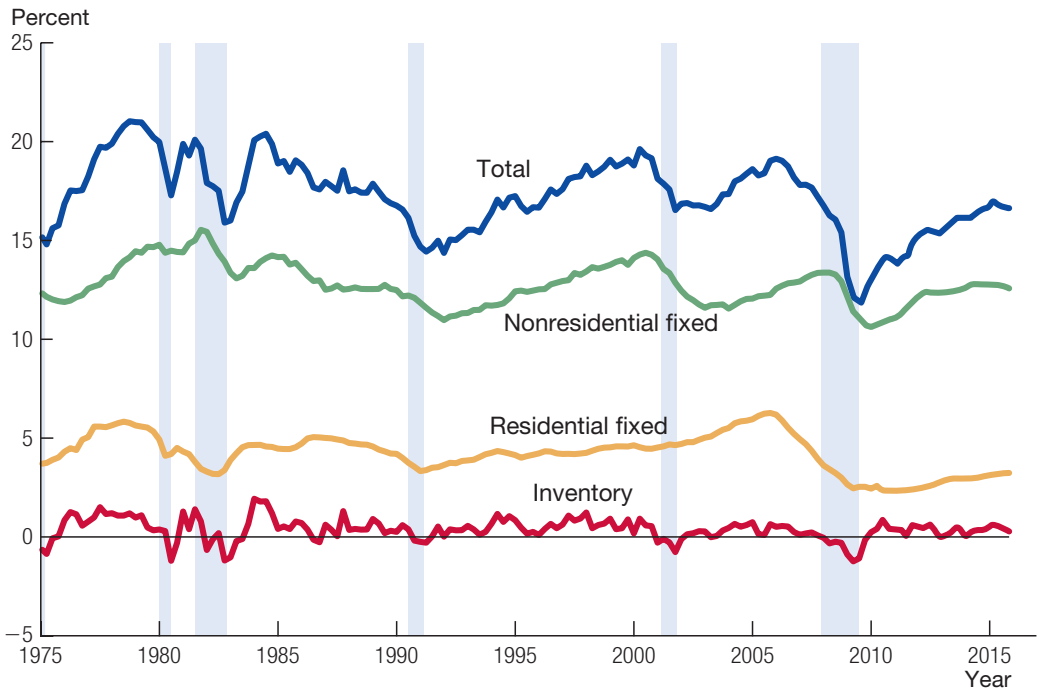
Alessandro Di Nola

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Investment over the business cycle

FIGURE 17.4

Components of Investment (as a Share of GDP)



Source: The FRED database.

Investment consists of three main components: nonresidential fixed investment, which includes equipment and structures purchased by businesses as well as intellectual property products; residential investment (housing); and the change in inventories held by businesses.

Source: Jones, Macroeconomics

Plan

1. Investment and capital accumulation in partial equilibrium
2. Dynamic general equilibrium with capital accumulation
3. Half-way mark for this course: taking stock and looking ahead

Readings and supplementary materials

1. Supplement on canvas: write-up of model including all the derivations
    - The additional notes on canvas often contain detailed derivations of the results shown in the lectures
2. Chapters 8 and 9 of Kurlat
    - general equilibrium model in his chapter 9 is similar but more complicated
    - also includes labor demand and supply which we abstract from

Investment and capital accumulation in partial equilibrium

Investment and capital accumulation in partial equilibrium

- Two time periods $t = 1, 2$, world ends afterwards
- Representative firm
- Firms produce using capital $K_t, t = 1, 2$

$$Y_t = A_t F(K_t), \quad F' > 0, F'' < 0$$

- K_1 is fixed
- K_2 can be changed through investment in period 1, I_1

$$K_2 = I_1 + (1 - \delta)K_1, \quad 0 < \delta \leq 1$$

where δ is depreciation (same equation as in Solow model)

Firm profit maximization

- Firm profits per period

$$\Pi_1 = A_1 F(K_1) - I_1, \quad \Pi_2 = A_2 F(K_2) + (1-\delta)K_2$$

- Why $(1-\delta)K_2$ in Π_2?

- Firms maximize present discounted value (PDV) of profits

$$W = \Pi_1 + \frac{\Pi_2}{1+r_1}$$

  where r_1 is interest rate between $t=1$ and $t=2$

- Why would firms maximize this present value? Why discounted at r_1?

- See supplement. In a nutshell because
  (a) firms are owned by households who can save and borrow at r_1
  (b) also firms can save and borrow at r_1

  (a)+(b) $\Rightarrow$ households instruct firms to maximize $W = \Pi_1 + \frac{\Pi_2}{1+r_1}$

Firm profit maximization

- Combining, firm problem is

$$W = \max_{K_2} \left\{ A_1 F(K_1) + (1-\delta)K_1 - K_2 + \frac{A_2 F(K_2) + (1-\delta)K_2}{1+r_1} \right\}$$

- Optimality condition

$$1 = \frac{A_2 F'(K_2) + 1 - \delta}{1 + r_1}$$

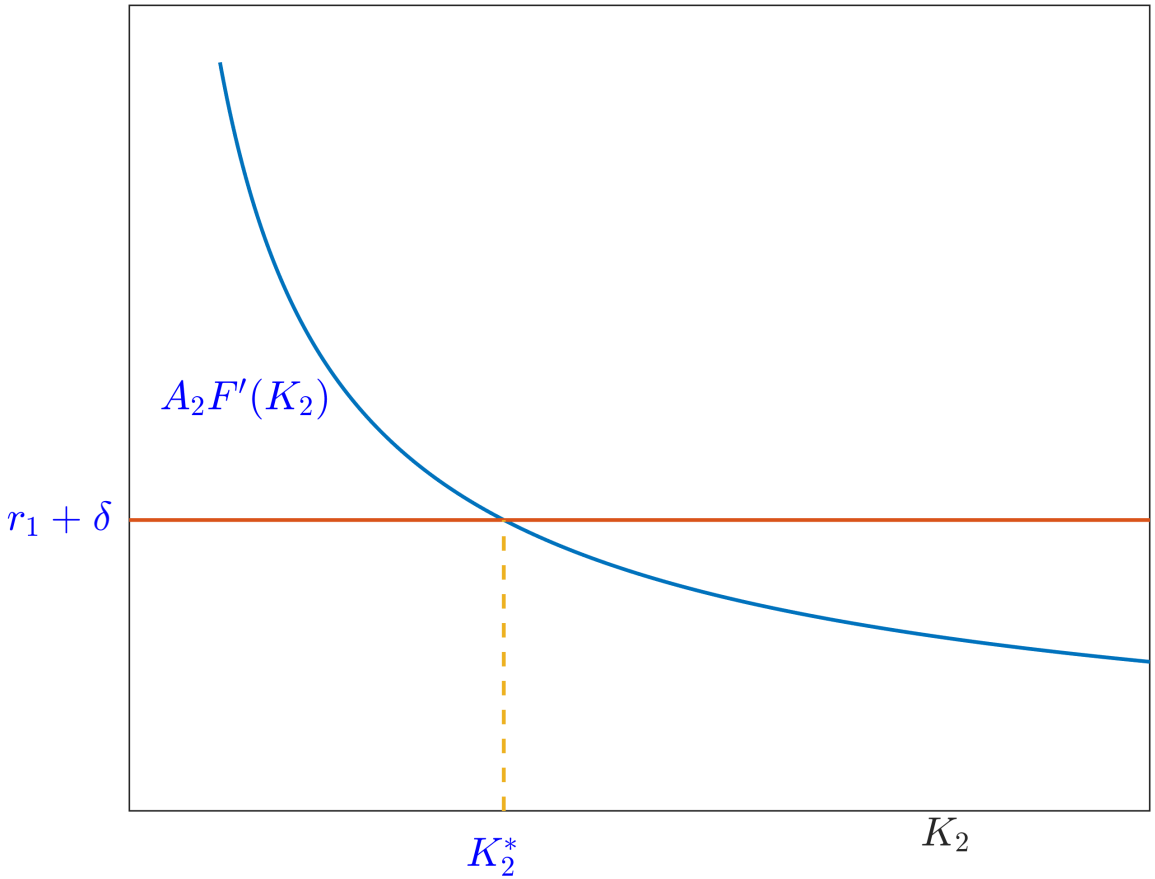
or

$$A_2 F'(K_2) = r_1 + \delta \qquad (*)$$

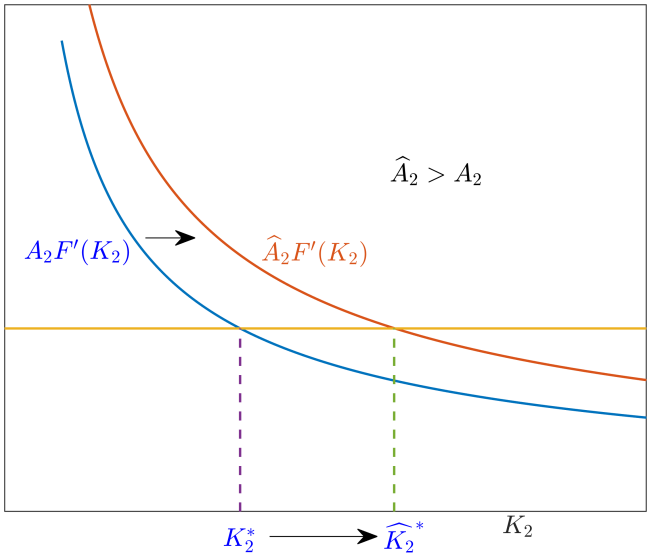
- $(*)$ pins down optimal capital K_2^* and therefore optimal investment

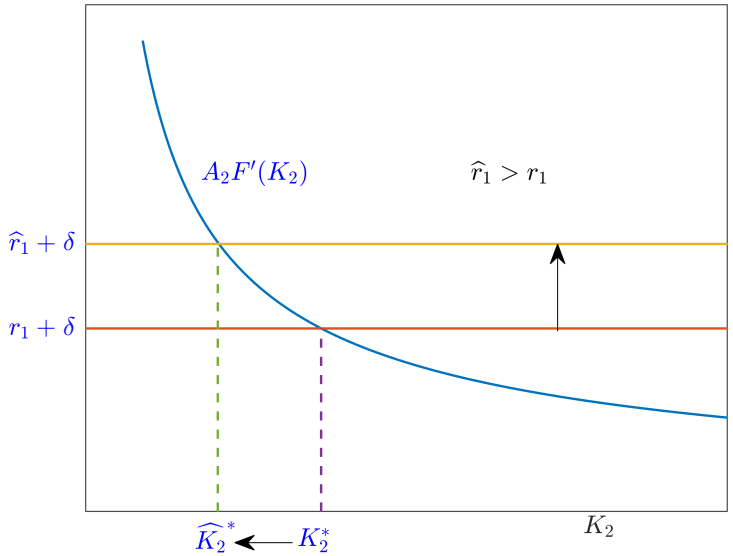
$$I_1^* = K_2^* - (1-\delta)K_1$$

Graphical representation of optimality condition



How does investment respond to changes in A_2 and r_1?





- What is the intuition for these comparative statics?

Capital and investment demand: parametric example

- Example: $F(K) = K^{\alpha}$ with $0 < \alpha < 1$
- Optimality condition

$$\alpha A_2 K_2^{\alpha-1} = r_1 + \delta$$

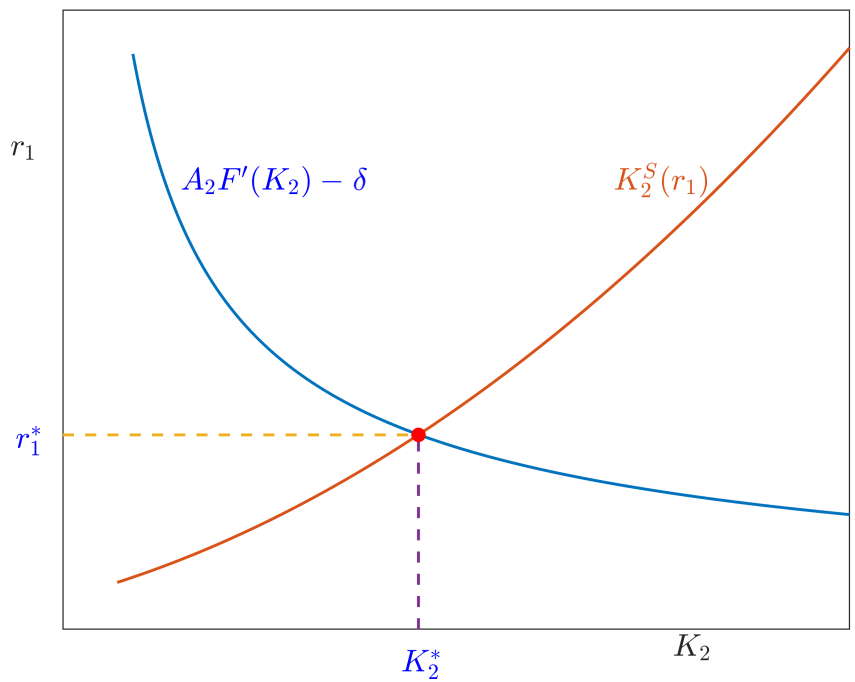
- $\Rightarrow$ optimal capital demand and investment

$$K_2^* = \left(\frac{\alpha A_2}{r_1 + \delta} \right)^{\frac{1}{1-\alpha}}$$

$$I_1^* = \left(\frac{\alpha A_2}{r_1 + \delta} \right)^{\frac{1}{1-\alpha}} - (1 - \delta) K_1$$

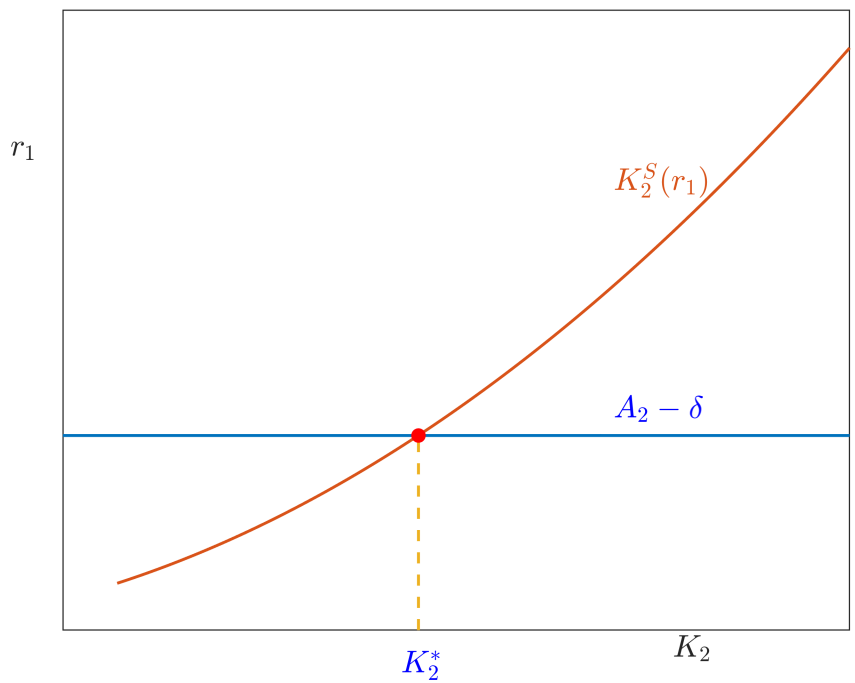
Warmup for part 2: equilibrium in the capital market

- Optimality condition $A_2 F'(K_2) = r_1 + \delta$ traces out a capital demand curve
- Plot of $r_1 = A_2 F'(K_2) - \delta$ vs K_2 is capital demand curve in (K_2, r_1)-space
- Warmup for part 2: can put this together w model of capital supply $K_2^S(r_1)$



Warmup: equilibrium with infinitely elastic capital demand

- Interesting special case: linear production $Y_2 = A_2 K_2$
- Optimality: $A_2 \gtrless r_1 + \delta \Rightarrow$ capital demand infinitely elastic at $r_1 = A_2 - \delta$
- Equilibrium interest rate $r_1 = A_2 - \delta$, quantity pinned down by supply



Dynamic general equilibrium with capital accumulation

Overview

- Lecture 3: household saving
- Lecture 4 so far: firm capital accumulation and production
- Now put these together in general equilibrium
- To make progress, work with special functional forms
  - no labor, linear production: $Y_t = A_t K_t$
  - full depreciation $\delta = 1$
  - utility with constant intertemporal elasticity of substitution
- Later: heavy use of these in our version of New Keynesian model
- Special assumptions but should be clear: construction of equilibrium follows steps that would also be valid with more general functions

Primitives of economy with capital accumulation

- **Preferences:** households have utility function

$$U(C_1) + \beta U(C_2) \quad \text{with} \quad U(C) = \frac{C^{1-\sigma} - 1}{1 - \sigma}, \quad \sigma \neq 1$$

  $\sigma > 0$: relative risk aversion; IES $= 1/\sigma$. For $\sigma = 1$, $U(C) = \log C$.

- **Technology:** firms have production function

$$Y_t = A_t K_t, \quad t = 1, 2$$

  and capital accumulates according to $K_2 = I_1 + (1 - \delta)K_1$ with $\delta = 1$, i.e.

$$K_2 = I_1$$

- **Resource constraints (feasibility):**

$$\text{goods in period 1:} \quad C_1 + I_1 = Y_1$$
$$\text{goods in period 2:} \quad C_2 = Y_2$$

Competitive equilibrium with capital accumulation

Definition: a competitive equilibrium are quantities $(C_1, C_2, I_1, K_2, Y_1, Y_2)$ and an interest rate r_1 such that

1. Utility maximization: taking as given r_1 and W, households choose (C_1, C_2) to solve

$$\max_{C_1, C_2} \; U(C_1) + \beta U(C_2) \quad \text{s.t.} \quad C_1 + \frac{C_2}{1 + r_1} = W$$

   where W is the PDV of firm profits (because households own firms)

2. Profit maximization: firms maximize $W = \Pi_1 + \frac{\Pi_2}{1+r_1}$ or equivalently

$$W = \max_{K_2} \left\{ A_1 K_1 - I_1 + \frac{A_2 K_2}{1 + r_1} \right\}, \quad K_2 = I_1, \; Y_1 = A_1 K_1, \; Y_2 = A_2 K_2$$

3. Market clearing: demand = supply for goods

$$\begin{aligned} \text{goods in period 1:} \quad & C_1 + I_1 = Y_1 \\ \text{goods in period 2:} \quad & C_2 = Y_2 \end{aligned}$$

Comment on market clearing conditions: credit market

- There really is a third market and correponding market clearing condition
  - credit market in which households and firms borrow/lend from/to each other at interest rate r_1 (recall that's why firms maximize PDV)
  - but can drop this due to Walras' Law
- Can write this as

$$b + B = 0$$

  where
  - b: household saving with $b < 0$ denoting borrowing
  - B: firm saving with $B < 0$ denoting borrowing
- Typical situation $b > 0$ and $B = -b < 0$, i.e. firms borrow from households to finance their investment
- See supplement. Can forget about this from now, no future appearances.

Solving for the competitive equilibrium allocation

- What is a convenient strategy to solve for the equilibrium allocation?

- Supplement: alternative strategy

▸ Derivation

Result: competitive equilibrium allocation (see supplement)

$$C_1 = \frac{\left(\frac{1}{\beta A_2}\right)^{1/\sigma} A_2}{1 + \left(\frac{1}{\beta A_2}\right)^{1/\sigma} A_2} A_1 K_1$$

$$C_2 = \frac{A_2}{1 + \left(\frac{1}{\beta A_2}\right)^{1/\sigma} A_2} A_1 K_1$$

$$K_2 = I_1 = \frac{1}{1 + \left(\frac{1}{\beta A_2}\right)^{1/\sigma} A_2} A_1 K_1$$

$$Y_1 = A_1 K_1$$

$$Y_2 = \frac{A_2}{1 + \left(\frac{1}{\beta A_2}\right)^{1/\sigma} A_2} A_1 K_1$$

$$1 + r_1 = A_2$$

Derivation: see supplement (one of the derivations, there are at least two)

Investment in the competitive equilibrium

- Focus on investment which is most interesting decision

$$I_1 = \frac{1}{1 + \left(\frac{1}{\beta A_2}\right)^{1/\sigma} A_2} A_1 K_1$$

- Note: can write this as saving rate $s(A_2)$ out of current output, Y_1:

$$I_1 = s(A_2) Y_1, \quad C_1 = [1 - s(A_2)] Y_1, \quad s(A_2) = \frac{1}{1 + \left(\frac{1}{\beta A_2}\right)^{1/\sigma} A_2}$$

- Contrast this with Solow model: $I_t = sY_t$ where $s =$ exogenously given

- In current model, saving rate is instead endogenous and depends on preferences (β, σ) and technology (A_2)!

- This is precisely what we mean when we say "Solow model is not microfounded but modern macro models are"

A boom due to an increase in A_1

- Assume current productivity $A_1 \uparrow$. What happens to $(C_1, C_2, I_1, Y_1, Y_2)$?
  - higher productivity means more output from the same capital
- Recall expressions for $(C_1, C_2, I_1, Y_1, Y_2)$ a few slides ago, e.g.

$$Y_1 = A_1 K_1, \qquad Y_2 = \frac{A_2}{1 + \left(\frac{1}{\beta A_2}\right)^{1/\sigma} A_2} A_1 K_1, \quad \ldots$$

- Clearly all of $(C_1, C_2, I_1, Y_1, Y_2)$ rise when A_1 rises
- Intuition: economy is more productive, people are richer $\Rightarrow$ increase consumption in both periods as well as investment
- Also note: both C_1 and I_1 rise. Reason: total resources Y_1 available for consuming or investing rise

$$C_1 + I_1 = Y_1, \quad Y_1 = A_1 K_1$$

A boom due to an increase in A_2

- Assume future productivity $A_2 \uparrow$. Current output $Y_1 = A_1 K_1$ is unchanged.
- $C_2 = Y_2$ unambiguously increases when A_2 rises

$$Y_2 = A_2 K_2 = \frac{1}{\frac{1}{A_2} + \left(\frac{1}{\beta A_2}\right)^{1/\sigma}} A_1 K_1 \uparrow \quad \text{when} \quad A_2 \uparrow.$$

- Effects on I_1 and C_1 depend on preferences: opposite signs unless both unchanged

$$I_1 = s(A_2) Y_1, \quad C_1 = [1 - s(A_2)] Y_1, \quad s(A_2) = \frac{1}{1 + \left(\frac{1}{\beta A_2}\right)^{1/\sigma} A_2}$$

- What happens to I_1 and C_1 depends on $1/\sigma =$ IES

$$\sigma > 1 \quad \Rightarrow \quad s'(A_2) < 0 \quad \Rightarrow \frac{\partial I_1}{\partial A_2} < 0, \quad \frac{\partial C_1}{\partial A_2} > 0 \quad \Rightarrow \quad I_1 \downarrow, \; C_1 \uparrow$$

- Why? And what is the intuition? (answer in next slide)
- From now on always assume $\sigma > 1$ (= also the empirically realistic case)

An increase in A_2: Intuition

- Under $\sigma > 1$, an increase in A_2 reduces investment I_1.
- With IES $= 1/\sigma < 1$, households strongly prefer to smooth consumption.
- Higher future productivity means more output per unit of investment.
- Households consume more today and invest less, while future consumption still rises.
- The income effect dominates the substitution effect, so investment falls despite its higher return.

A boom due to increases in both A_1 and A_2

Can also consider combination: both A_1 and A_2 rise at the same time

- For example, A_1 rises and A_2 is positively correlated with A_1

$$\log A_2 = \rho \log A_1 + \varepsilon_2$$

  where $\rho > 0$ captures persistence, holding ε_2 fixed

- Output and consumption rise in both periods
- Under $\sigma > 1$, investment is ambiguous: higher A_1 raises I_1, while higher A_2 lowers it
- Such correlated productivity movements will be important in next lecture

Oil shocks as productivity shocks (or gas shocks)

- What on earth is a drop in productivity?
  - do we think people forget how to make stuff? Not really
  - hard to come up with sensible justifications
- One possible justification: oil shocks or energy (gas etc) shocks
- Technology: firms use oil to produce

$$\tilde{Y}_t = \tilde{A}_t K_t^{\alpha} O_t^{1-\alpha}.$$

- Firms maximize output net of oil expenditure

$$Y_t = \max_{O_t} \ \tilde{A}_t K_t^{\alpha} O_t^{1-\alpha} - p_t O_t \quad \text{where } p_t = \text{oil price}$$

$$\Rightarrow \quad Y_t = A_t K_t \ \text{ where } \ A_t = \text{effective productivity} = \alpha \tilde{A}_t^{1/\alpha} \left(\frac{1-\alpha}{p_t} \right)^{(1-\alpha)/\alpha}$$

so an increase in p_t causes a drop in effective productivity

Half-way mark for this course: taking stock and looking ahead

What we have done in first half of course

- Learned the basics of modern macroeconomics
- What is a model? "The map is not the territory" and so on
- Reminded ourselves of basic microeconomics
  - intratemporal choice, e.g. labor supply
  - intertemporal choice, e.g. consumption, saving, investment
- Learned the following key concepts
  1. Competitive equilibrium
  2. Pareto effiency
  3. Welfare theorems

What we have done in first half of course

- Worked with some key parameters, corresponding functional forms
  - elasticity of substitution in production
  - intertemporal elasticity of substitution
- Reminded ourselves that income and substitution effects are everywhere!
- Applied these tools to think about some key topics
  1. Labor supply and labor demand, long-run trends in hours worked
  2. Substitution in production, e.g. in response to cut-off of Russian gas
  3. Consumption and saving decisions, permanent income hypothesis & its shortcomings, MPCs out of transitory & permanent income shocks
  4. Investment and capital accumulation

What we will do in second half of course

- Put your basic training to use to think about more and more applied topics
  - Business Cycle Macro and Lucas Critique
  - New Keynesian Model I
  - New Keynesian Model II
  - The Financial Crisis, Asset Bubbles
  - Unemployment (Pissarides), Inequality in Macro
- Looking for something to read during reading week? Read Mankiw and Weinzierl (2011) "An Exploration of Optimal Stabilization Policy" `https://www.brookings.edu/wp-content/uploads/2011/03/2011a_bpea_mankiw.pdf`
  - Two-period version of New Keynesian model I will teach is based on this paper
  - Model from this lecture but with "nominal rigidities", i.e. sticky prices
  - You have basic tools to read this but may still lack some vocabulary
  - Read sections I-IV and IX-X (skip sections V-VIII)

Appendix: Equilibrium and firm value

- A competitive equilibrium consists of quantities $(C_1, C_2, I_1, K_2, Y_1, Y_2)$ and an interest rate r_1 satisfying household utility maximization, firm profit maximization, and market clearing.

- Firm optimality gives $1 + r_1 = A_2$; full depreciation gives $I_1 = K_2$.

- Substitute into the present discounted value of profits:

$$\begin{aligned} W &= A_1 K_1 - I_1 + \frac{A_2 K_2}{1 + r_1} \\ &= A_1 K_1 - K_2 + K_2 = A_1 K_1. \end{aligned}$$

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Appendix: Seven equations

The seven endogenous variables $(C_1, C_2, I_1, K_2, Y_1, Y_2, r_1)$ are determined by seven independent equations:

$$\begin{aligned}
\text{Euler equation:} \quad & \frac{C_2}{C_1} = [\beta(1 + r_1)]^{1/\sigma} \\
\text{Household budget:} \quad & C_1 + \frac{C_2}{1 + r_1} = A_1 K_1 \\
\text{Firm optimality:} \quad & 1 + r_1 = A_2 \\
\text{Capital accumulation:} \quad & I_1 = K_2 \\
\text{Period-1 goods clearing:} \quad & C_1 + I_1 = Y_1 \\
\text{Period-1 production:} \quad & Y_1 = A_1 K_1 \\
\text{Period-2 production:} \quad & Y_2 = A_2 K_2
\end{aligned}$$

Second-period goods-market clearing, $C_2 = Y_2$, follows from this system.

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Appendix: Reduce to one unknown

- Substitute $1 + r_1 = A_2$ into the Euler equation and define

$$\frac{C_2}{C_1} = x(A_2), \qquad x(A_2) = (\beta A_2)^{1/\sigma}.$$

- First-period goods clearing and production give $C_1 = A_1 K_1 - I_1$.
- The household budget, $I_1 = K_2$, and production imply

$$C_2 = A_2(A_1 K_1 - C_1) = A_2 I_1 = Y_2.$$

- Substitute into the Euler equation:

$$\frac{A_2 I_1}{A_1 K_1 - I_1} = x(A_2).$$

  Only I_1 remains unknown!

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Appendix: Recover the allocation

- Rearrange the equation for investment:

$$[A_2 + x(A_2)]I_1 = x(A_2)A_1K_1,$$

$$I_1 = \frac{x(A_2)}{A_2 + x(A_2)}A_1K_1.$$

- Recover capital, current output, and current consumption:

$$K_2 = I_1, \quad Y_1 = A_1K_1, \quad C_1 = Y_1 - I_1.$$

- Then recover future output, future consumption, and the interest rate:

$$Y_2 = A_2K_2, \quad C_2 = Y_2, \quad r_1 = A_2 - 1.$$

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